

Spatial Disorder Can Enhance Temporal Order in a Hierarchical Active Matter

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Abstract

In active matter, self-propelled particles typically generate collective dynamics through interactions. In many experimental systems, these particles achieve self-propulsion via more microscopic active units, such as cilia and flagella. However, how these active units contribute to the collective dynamics of larger hierarchical particles remains under-explored. Here, we examine active-agent-laden droplets as a model system. These droplets achieve autonomous propulsion and rotation due to neutral swimmers on their surfaces. Using hydrodynamic simulations, we analyze their dynamics in free space and near a no-slip wall. For a single droplet, we observe a petal-like circular motion in its trajectory. For an array of droplets in equilibrium positions, we identify several collective rotational modes, including spontaneous reorientation, phase-locking, and synchronization. These phenomena arise from translation-rotation hydrodynamic coupling. Remarkably, when the equilibrium positions of these droplets are perturbed by a designated displacement field, their dynamics can shift from a less ordered to a more ordered collective mode. Our findings establish a new paradigm where mesoscale active carriers inherit and amplify microscopic actuation through surface-active flows and hydrodynamic interactions, offering design principles for programmable active matter systems.

Keywords: Hydrodynamics, Synchronization, Droplets, Active Matter, Microswimmers

1 Introduction

Collective dynamics, a hallmark of active matter, encompasses a range of emergent phenomena in which a global order or a large-scale coherent structure can arise, despite the absence of such order at the individual-unit level [1, 2]. This order can manifest in various forms, including spatial patterns such as motility-induced phase separation [3], directional behaviors like swarming [4], flocking [5], and spontaneous flows [6], as well as temporal ordering, such as synchronization [7–12]. These emergent orders play a crucial role in living systems. Take the temporal order as an example: the synchronized beating of cilia is essential for the swimming of ciliated microorganisms [13–16] and for the effective pumping of the mucus flow over an airway epithelium [17–19].

Living active matter typically exhibits hierarchical structure. For example, motor proteins can drive the active rotation of individual cilia, which can then beat collectively to mobilize the ciliated cells, allowing for their coordinated movements [15, 20, 21]. Current research in active matter primarily focuses on a simple two-level scenario, that is, examining how active units as a whole generate collective dynamics. However, it remains unclear how the activity of the basic units can ascend the hierarchy and dictate the emergent order at the higher levels of the system. Recently, experimental steps have been taken to bridge this gap by demonstrating how microscopic active agents can actuate mesoscale carriers, such as by hydrodynamically powering the rotation of symmetric structures [22], inducing large

shape deformations in giant lipid vesicles [23], or driving the locomotion of deformable enclosures [24]. Related experiments have further shown that interfacial bacterial biofilms can remodel oil droplets through defect-mediated buckling and tubulation [25], while motile bacteria within membraneless droplets can give rise to activity-enhanced droplet transport and anti-phase entrainment of interface fluctuations between neighboring droplets [26]. However, these experimental systems are typically driven by the stochastic, collective propulsion of numerous confined active units, generating chaotic flows or complex interfacial deformations that obscure the fundamental physical mechanisms of energy transfer across scales. Furthermore, while the autonomous dynamics of these isolated carriers and local pairwise coordination of their interfacial fluctuations have been observed, it remains largely unexplored whether arrays of such hierarchically driven structures can hydrodynamically couple to establish long-range temporal order, such as synchronization and phase-locking, and crucially, whether these emergent collective states can be rationally controlled.

To address this open question, we use hydrodynamic simulations to study a model hierarchical active matter system—a collection of droplets laden with active agents. In this system, a neutral swimmer self-propels at the interface of a droplet, rendering the droplet active through the fluid flow generated in the continuous phase by the active interfacial flow. An array of such non-equilibrium droplets can then interact hydrodynamically with one another and with nearby boundaries, giving rise to re-orientations and coordinated motions. We focus on analyzing the dynamics of these active droplets.

We adopt the Rigid Multiblob Method (RMM) [27, 28], which is a Stokesian dynamics framework particularly designed for modeling rigid particles of arbitrary shape moving in a Stokes flow actuated by external forces and interfacial flows. As a benchmark, our simulation of a no-slip sphere in an unconfined Stokes flow shows excellent agreement with the analytical theory [29], confirming the quantitative accuracy of our model (Supplementary Information Section 1).

Next, we use RMM to systematically investigate how a neutral swimmer confined to a droplet interface impacts both single-droplet dynamics and the collective dynamics of multiple such droplets. For a single droplet near a no-slip wall, we observe a petal-like orbital trajectory. When the droplet is further optically constrained near an equilibrium position, the steady-state trajectory of the active agent can be reoriented by the nearby wall, causing the droplet to spin about the wall-normal direction. We argue that this behavior arises from a translational-rotational hydrodynamic coupling between the droplet and the wall.

More interestingly, when an array of such droplets dispersed in an unconfined domain are trapped in potential wells, they can develop various temporal orders in their active agents—including synchronization, phase-locking, and reorientation—depending on the droplet number and the well stiffness. While the droplets self-rotate, their centers of mass also exhibit coordinated oscillations about their equilibrium positions. Inspired by these coordinated oscillations, we impose a displacement field to the equilibrium positions of the droplets according to the oscillation pattern and find that the dynamic modes of the droplet array can be tuned between a less ordered mode and a more ordered one. We explain our findings using a theoretical model. Together, we provide new insights into synchronization dynamics in hierarchical active matter and propose using displacement noise to control their collective spatiotemporal dynamics.

2 Results

2.1 Single droplet: locomotion and spontaneous reorientation near a no-slip wall

We first use RMM to investigate the hydrodynamic behavior of a single droplet laden with a neutral swimmer at its interface (Fig. 1(a)). More specifically, we consider two configurations: (i) free space (unbounded Stokes flow) and (ii) partially confined geometry with an infinite no-slip substrate.

We examine configuration (i) as a benchmark study. Specifically, we consider a spherical droplet of radius R with a single active agent attached to its interface. The agent is modeled as a source dipole of strength v_s pointing within the tangent plane to the droplet and generating an interfacial flow. Such flow can further invoke a three-dimensional (3D) bulk flow (Fig. 1(a) and Fig. S1(e)), which in turn advects the droplet. Because of the symmetry of the spherical droplet, we choose a coordinate system such that the agent is initially placed on the equatorial plane of the droplet and points clockwise in a top view (i.e., viewed from the $+z$ -direction, i.e., the north pole, see Fig. 1(a)). The momentum exchanged between the droplet and the surrounding continuous phase via the interfacial flow gives rise to a hierarchy of hydrodynamic multipoles. The two lowest-order contributions are the Stokeslet (point force) and the rotlet (point torque). If, to leading order, we idealize the droplet's hydrodynamic signature as the superposition of a point force and a point torque, the observed circular trajectory of the active droplet can be rationalized as follows: the droplet's center of mass (COM) self-propels with a linear velocity U_0 while simultaneously self-spinning with an angular velocity Ω_0 , and the radius of the resulting circular orbit is given by $r_{\text{orb}} \equiv U_0/\Omega_0$. The handedness of the trajectory is set by the initial orientation of the microswimmer. By entering the slip boundary condition into the exact relations derived from the reciprocal theorem [31],

$$\mathbf{U}_0 = -\frac{1}{4\pi R^2} \int_S \boldsymbol{\nu} dS, \quad \text{and} \quad \boldsymbol{\Omega}_0 = -\frac{3}{8\pi R^3} \int_S \mathbf{n} \times \boldsymbol{\nu} dS, \quad (1)$$

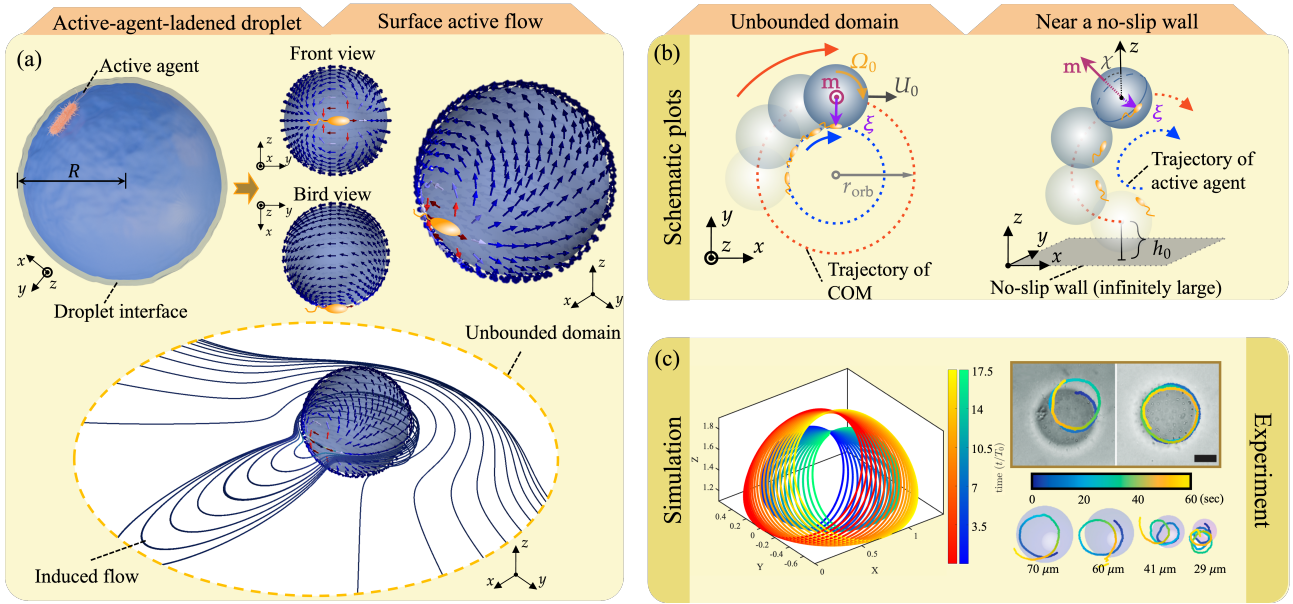


Fig. 1 Dynamics of a single untethered droplet. (a) Top left: illustration of a microswimmer or agent (orange ellipsoid) moving on the interface of a droplet (blue sphere); top middle and right: agent-induced active interfacial flow field (arrows); bottom: streamlines of the slaved bulk flow generated by the interfacial flow in an unbounded space. (b) Left: a single untethered droplet in an unbounded domain exhibits combined translational and rotational motion along a circular trajectory (dashed orange line). The corresponding trajectory of the active agent is denoted by a dashed blue line. Right: a single free droplet near a no-slip boundary exhibits petal-like orbital motion. The unit vectors $\mathbf{m}$ and $\boldsymbol{\xi}$ correspond to the droplet's rotational axis and the positional vector of the agent from the droplet center, respectively, and χ is the pitch angle between $\mathbf{m}$ and the z -axis. (c) Left: 3D trajectories of the center-of-mass of the droplet and the agent. The color mapping indicates the dimensionless time t/T_0 , where the warm gradient (red-to-yellow) represents the COM trajectory, and the cool gradient (blue-to-green) traces the active agent trajectory. Right: clockwise rotation (left) and counter-clockwise rotation (right) of droplets, as illustrated by the trajectories of PS particles; Lower parts are representative trajectories of PS particles adhered to the surfaces of dodecane droplets with diameters of 29, 41, 60, and 70 μm , suspended in a 10 g L $^{-1}$ potassium nitrate solution in water [30]. The scale bar is 20 μm . Color indicates absolute time in seconds.

one can calculate the resulting linear velocity $\mathbf{U}_0$ and angular velocity $\boldsymbol{\Omega}_0$ of the droplet. Here, R is the radius of the droplet, S denotes its surface, $\mathbf{n}$ is the outward surface normal vector, and $\boldsymbol{\nu}$ represents the active surface slip velocity. The measured U_0 and Ω_0 in our simulation match well with this calculation, further verifying the accuracy of our method (Supplementary Information Section 1).

Because the viscous drag coefficients for the translational and rotational motion of a spherical particle in an unbounded space scale as R and R^3 , respectively, the two velocity magnitudes should obey $U_0 \propto v_s/R$ and $\Omega_0 \propto v_s/R^3$, respectively [31]. These scaling behaviors are well reproduced by our simulation (Supplementary Information Section 1 and Fig. S1), once again confirming the accuracy of our numerical method. Note that circular (chiral) motion is also observed in a bacteria-driven droplet experiment (Fig. 1(c)), where the droplet's angular speed scales inversely with its size [30]. In that experiment, the droplet was propelled by multiple bacteria on its interface, which is in contrast to our single-agent model. The activity of the bacteria-driven droplet should roughly scale with the droplet's surface area, i.e., $\propto R^2$. As a result, the angular speed follows $R^2 \cdot R^{-3} = R^{-1}$ —this explains the different scaling exponents between the bacteria-droplet system and our single-agent model. For a similar reason, this inverse scaling law holds in another experiment where solid symmetric micro-gears are powered by the hydrodynamic torque dipoles of multiple confined *E. coli* bacteria [22].

Next, we examine wall effects by studying configuration (ii). The wall is positioned at the $z = 0$ plane with a no-slip boundary condition, and the agent is initially ($t = 0$) positioned on the equator of the droplet. We vary two geometric parameters: (a) the initial pitch angle of the agent, $\chi(t = 0) \equiv \chi_0 \in [0, \pi/2]$, defined as the angle between the instantaneous rotational axis $\mathbf{m}$ of the droplet and the z -axis (Fig. 1(b)), and (b) the initial height of the droplet's COM, $\bar{h}_0 \equiv h_0/R$. We restrict $\chi_0 \in [0, \pi/2]$ because a $\chi_0 > \pi/2$ scenario can be mapped to a $\chi_0 < \pi/2$ case when the agent moves to the other side of the droplet in a later time.

We first discuss the two trivial scenarios, namely $\chi_0 = 0$ and $\pi/2$. When the initial pitch angle is close to zero ($\chi_0 \approx 0$), the droplet trajectory remains approximately within the xy -plane with small vertical oscillations (Fig. S2(a)), which is similar to the circular trajectory observed in the unbounded configuration (i). When $\chi_0 = \pi/2$, the resulting motion is a stable trochoidal trajectory, which can be understood as a superposition of a circular orbit within a vertical plane and a constant translational drift parallel to the wall (Fig. S2(c)).

For nontrivial initial pitch angles $0 < \chi_0 < \pi/2$, the droplet exhibits complex petal-like trajectories (Fig. 1(c) and Fig. S2). This behavior contrasts with persistent random walks observed in active droplets encapsulated with multiple bacteria, the collective motion of which generates chaotic-like internal flows, agitating the droplet in a random manner [33]. Here, the petal-like trajectory can be regarded as a superposition of a self-spinning and a precession-like motion. To clarify its underlying mechanism, we decompose the prescribed surface-active-flow slip into generalized squirmer modes [32], as detailed in Supplementary Information Section 3. The decomposition identifies the first-order modes $\tilde{B}_{1,1}$ and $C_{0,1}$ as setting the intrinsic translational velocity $\mathbf{U}_0$ and angular velocity $\boldsymbol{\Omega}_0$, respectively, while the

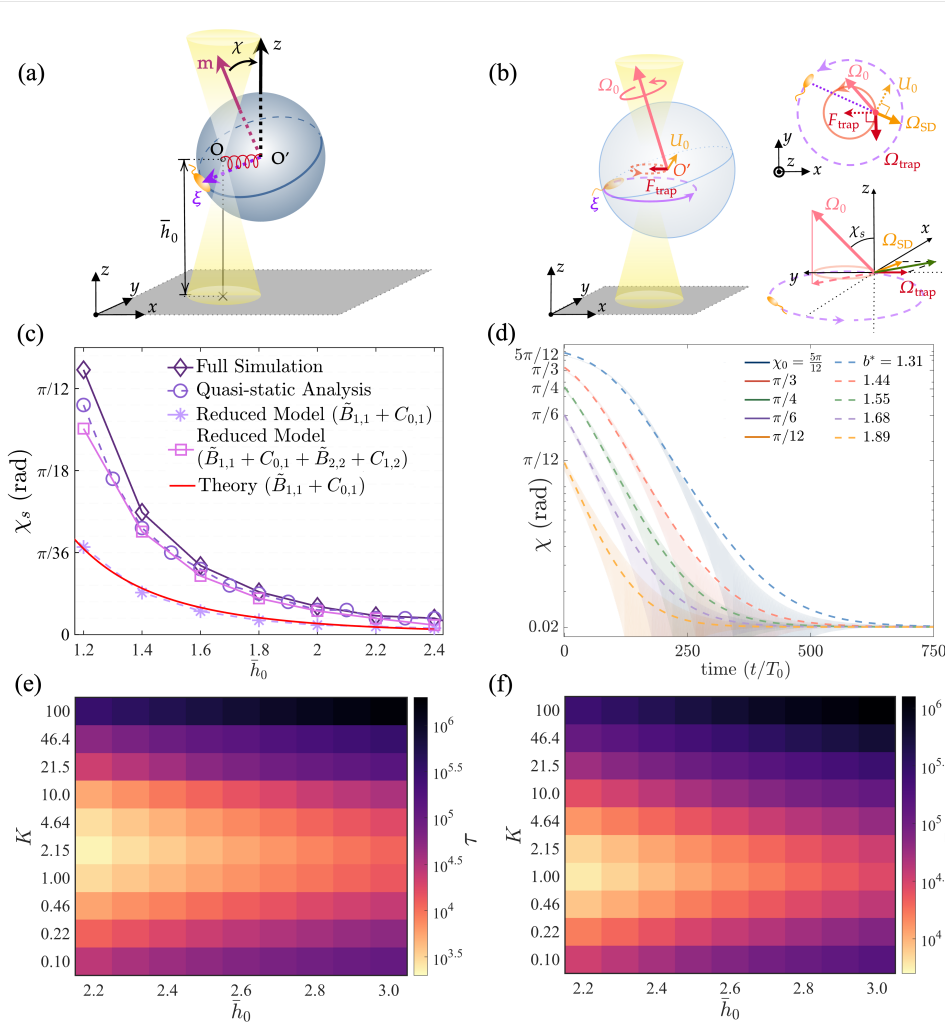


Fig. 2 Re-orientation of a single tethered droplet. (a) Schematic of a single droplet constrained by an optical trap (illustrated by yellow cones) being suspended near a no-slip wall in the bottom. Here, h_0 denotes the height of the trap center, ξ represents the positional vector of the active agent with respect to the droplet center, $\mathbf{m}$ indicates the rotational axis, and χ is the angle between $\mathbf{m}$ and the z -axis. (b) The left panel presents a three-dimensional (x - y - z) side view of the active droplet confined by an optical trap above a no-slip bottom wall. The top-right and bottom-right panels display the corresponding top (x - y plane) and side (y - z plane) views, respectively. The purple and red dashed lines trace the stable orbital trajectories of the active agent and the droplet's COM, respectively. Vector $\mathbf{\Omega}_0$ represents the intrinsic rotational angular velocity, $\mathbf{U}_0$ is the intrinsic translational propulsion velocity generated by the active agent, and $\mathbf{F}_{\text{trap}}$ is the restoring force from the optical trap. During counter-clockwise motion, the $\mathbf{F}_{\text{trap}}$ the agent's equivalent source dipole interact with the bottom wall, inducing angular velocities denoted as $\mathbf{\Omega}_{\text{trap}}$ and $\mathbf{\Omega}_{\text{SD}}$, respectively. To maintain angular velocity balance within the x - y plane as it is in the steady state circular trajectory, $\mathbf{\Omega}_0$ must participate to offset $\mathbf{\Omega}_{\text{trap}}$ and $\mathbf{\Omega}_{\text{SD}}$. This coupling results in a finite steady-state pitch angle χ_s and dictates that the agent's trajectory plane remains consistently below the COM's trajectory plane. This physical mechanism applies identically to clockwise orbital motion. (detailed analysis in Supplementary Information Section 5) (c) Steady-state pitch angle χ_s as a function of the dimensionless trap height $\bar{h}_0$ at $K = 10$. Diamonds show the full simulations and circles the corresponding quasi-static analysis (see Fig. S6f). A reduced squirmer model retaining the $\bar{B}_{1,1}$ and $C_{0,1}$ modes captures the overall decrease of χ_s with increasing $\bar{h}_0$ (asterisks), while the extended model including the additional $\bar{B}_{2,2}$ and $C_{1,2}$ modes (squares) gives substantially closer agreement with the full simulations. The red solid line shows the analytical prediction for the $\bar{B}_{1,1}$ and $C_{0,1}$ reduced model, obtained from the wall-induced angular velocities of the corresponding squirmer modes [32?] (detailed derivations in Supplementary Information Section 7). (d) Relaxation of the pitch angle χ from different initial orientations χ_0 at $K = 10$ and $\bar{h}_0 = 2.2$. All trajectories converge to a common steady-state angle χ_s . Dashed curves are fits to $\chi(t) = \chi_s + \arctan[A \exp(-bt)]$, with a common steady-state angle χ_s and fitted amplitude A and relaxation rate b . The corresponding relaxation rates $b^* = b \times 10^4$ are given in the legend. (e) Relaxation time $\tau \equiv 1/b$ for $\bar{h}_0 \in [2.2, 3.0]$ in full simulation. Color represents the relaxation time. (f) Relaxation time τ for $\bar{h}_0 \in [2.3, 3.0]$, using the theory in equation 3.

higher-order $\bar{B}_{2,2}$ and $C_{1,2}$ modes characterize the leading non-axisymmetric components of the disturbance flow. In an unbounded fluid, the leading translational and rotational squirmer modes propel the force- and torque-free droplet perpendicular to its intrinsic rotation axis, producing the circular motion described in configuration (i). Near a no-slip wall, reflections of the slip-generated disturbance flow produce an additional wall induced hydrodynamic angular response. In particular, the leading source-dipole contribution generates a wall-induced angular velocity transverse to the instantaneous propulsion direction, causing the rotational axis $\mathbf{m}$, and hence the plane of the intrinsic circular orbit, to undergo a slow precession. The superposition of this wall-induced precession and the fast intrinsic circular swimming produces the intricate, highly periodic petal-like trajectory. This trajectory is successfully reproduced in a reduced model, in which the agent's hydrodynamic effect is represented by a reduced squirmer model retaining the leading slip modes source dipole $\bar{B}_{1,1}$ and rotlet $C_{0,1}$ (Supplementary Information Section 3 and Fig.S2(d)). In our simulation, a gradual increase of χ_0 from 0 to $\pi/2$ drives a continuous transition from circular to petal-like, and

finally to trochoidal trajectories (Fig. 1(c), Fig. S2(b,c)). A quantitative analysis of the trajectory mean radius of curvature r_c and its scaling with $\bar{h}_0$ and χ_0 are provided in Supplementary Information Section 2.

Next, in our simulation, we attach a harmonic spring to the droplet to mimic many experimental systems, where optical tweezers are used to constrain and manipulate active droplets [34, 35]. Specifically, we apply a spring force $\mathbf{F}_{\text{trap}} = -K(\mathbf{O}' - \mathbf{O})$ to the droplet's COM located at $\mathbf{O}'$ with its equilibrium position $\mathbf{O}$ (Fig. 2(a)), where K is the spring constant. This spring force can alternatively be interpreted through biochemical mechanisms, such as phototactic response to a fixed light source or chemotactic attraction to a local nutrient reservoir.

Similar to the untethered droplet discussed in Fig. 1(b), we find that the tethered droplet in configuration (i) also undergoes a circular motion (Supplementary Information Section 5 and Fig. S3). From the steady-state velocity balance in the orbital plane, we have derived the orbital radius r_{orb} of the tethered droplet as a function of K (Supplementary Information Section 5):

$$r_{\text{orb}}(K) = \frac{\gamma_t U_0}{\sqrt{K^2 + (\gamma_t \Omega_0)^2}}, \quad (2)$$

where $\gamma_t = 6\pi\eta R$ is the translational drag coefficient. This relationship indicates that the orbit is determined by the interplay between active propulsion $\gamma_t U_0$, trap stiffness K , and viscous force $\gamma_t \Omega_0$. The trivial case $r_{\text{orb}}(K = 0) = U_0/\Omega_0$ is exactly the result for a free droplet discussed in Section 2.1. As the trap stiffness K increases, the orbit is more constrained by the spring's restoring force, leading to a smaller r_{orb} . Note that our constrained active-agent-laden droplets exhibit some similarity to certain cilia models, where discrete ciliary beating is abstracted as orbiting point particles [8, 13, 36] or spring-tethered active particles [35, 37, 38]. However, our system represents a hierarchical active matter in which the active agent orbits along the droplet's interface rather than being directly attached to a spring.

We further complicate the above system by introducing a no-slip wall near the droplet. The presence of the wall breaks the rotational symmetry of the problem. Indeed, our simulation finds that the rotational axis $\mathbf{m}$ of the agent tends to align towards the wall-normal direction, i.e., the z -axis (Fig. 2(b)). In the long-time limit, when the dynamic droplet reaches a steady state, its pitch angle $\chi(t)$ decreases to a considerably small value $\chi(t \rightarrow \infty) = \chi_s$ (Fig. 2(c,d)). In the wall-normal (z) direction, any oscillation of the droplet will be counter-balanced by the spring force and damped by the viscous force. At minimum dissipation, the droplet's COM will remain to be within the xy plane. Indeed, the steady-state trajectory of the droplet is a circle in the xy plane. As a result, the angular momentum of the droplet should be along the z -axis, resulting in a small steady-state pitch angle χ_s for the agent. We further perform quasi-static force-torque analysis and find that all initial pitch configurations converge to the alignment state, giving rise to small χ_s 's (Fig. 2(c), S3(b)). In contrast, a free (untethered) active droplet with $K = 0$ lacks the trap-mediated restoring force to facilitate dissipation of the vertical oscillation of the droplet, leading to a petal-like trajectory (Fig. 1(c)). The amplitude of these oscillations diminishing as the initial pitch angle increases (Fig. S2(f)). This spontaneous reorientation underscores the distinct nature of our system as a hierarchical active matter. Unlike simple active particles directly attached to a spring, the active agents here orbit along the droplet's interface, inducing a translation-rotation coupling that is essential for the observed alignment effect.

As illustrated in Fig. 2(c), the steady-state pitch angle χ_s is closer to 0 as the trap height $\bar{h}_0$ increases. When the droplet is confined close to the boundary ($\bar{h}_0 \lesssim 2.2$), the asymmetric hydrodynamic repulsion generated by the active agent breaks the rotational symmetry. This geometric constraint counteracts the reorientation tendency, stabilizing the droplet to a finite steady-state pitch angle ($\chi_s > 0$). As the separation distance increases, this wall-induced asymmetry diminishes; consequently, for $\bar{h}_0 \gtrsim 2.2$, the confinement-mediated reorientation dominates, driving χ_s asymptotically toward zero. The equilibrium orientation is thus governed by a competition between the optical trap's restoring constraint and the height-dependent wall interactions. This physical picture is quantitatively supported by the excellent agreement between our dynamic simulations and the quasi-static force-torque analysis, which accurately captures the monotonic decay of χ_s from $\approx \pi/12$ to 0 as $\bar{h}_0$ increases from 1.2 to 2.4.

The reorientation relaxation time τ of the constrained droplet in the above scenario depends on three key parameters: initial pitch angle χ_0 , initial height $\bar{h}_0$, and spring constant K . By fitting the pitch angle with $\chi(t) = \chi_s + \arctan[A \exp(-bt)]$, we can measure the relaxation time $\tau = 1/b$. Our data show that τ is insensitive to the initial pitch angle χ_0 (Fig. 2(f)). Thus, we plot τ as a function of $\bar{h}_0$ and K in Fig. 2(g,h). Our theoretical model (Supplementary Information Section 7) predicts that the reorientation rate τ^{-1} scales with the trap stiffness and the orbital radius as $\tau^{-1} \propto K[r_{\text{orb}}(K)]^2$. Substituting equation 2, we obtain the analytical expression for the relaxation time:

$$\tau(K, h_0) = \frac{8h_0^5 [K^2 + (6\pi\eta R\Omega_0)^2]}{27\pi\eta R^4 U_0^2 K}. \quad (3)$$

This scaling law, plotted in Fig. 2(h), captures the non-monotonic dependence on the trap stiffness K as observed in the simulations. Clearly, there exists an optimal stiffness $K_{\text{opt}} \approx \gamma_t \Omega_0$ that minimizes τ at a fixed $\bar{h}_0$. In other words, the relaxation exhibits a non-monotonic behavior with respect to K : both weak and strong confinement regimes show significantly large τ compared to that at the optimized, intermediate K . For weak confinement regime, insufficient droplet-wall coupling weakens the alignment effect due to the translation-rotation coupling, allowing the system to stay longer with the petal-like trajectory. For strong confinement regime, overly rigid constraints drive rapid convergence to small-amplitude limit cycle trajectories before wall-mediated reorientation dynamics can be fully

developed. These simulation insights uncover a design principle for experimental systems, suggesting that precisely tuned confinement strength can maximize the reorientation efficiency through balanced fluid-structure interactions.

2.2 Multi-droplet array: synchronization, phase-locking and reorientation

In the following sections, we consider an array of constrained droplets in configuration (i), with each of them being subjected to a harmonic spring tethered to its equilibrium position. Thermal fluctuations at room temperature (300 K) are included as random noise (Supplementary Information Section 8) [28, 39, 40]. We consider N droplets with their initial centroid positions uniformly spaced along the z -axis at intervals $l_{i,i+1} \equiv l$ for $\forall i \in \{1, 2, \dots, N-1\}$, as illustrated in Fig. 3(a). The phase ϕ_i of active agent i (i.e., the agent on the i 'th droplet) is defined as a cumulative rotation angle about its instantaneous rotational axis $\mathbf{m}_i$, modulo 2π . We express this through the time-integrated relation [37]:

$$\phi_i(t) = \left(\int_0^t \boldsymbol{\omega}_i(\tau) \cdot \mathbf{m}_i d\tau \right) \bmod 2\pi, \quad (4)$$

where $\boldsymbol{\omega}_i$ represents the angular velocity vector. The inter-droplet phase difference is subsequently defined by $\delta\phi_{i,i+1} \equiv |\phi_{i+1} - \phi_i| \in [0, 2\pi)$, which characterizes the orientational coordination or correlation between adjacent active agents i and $i+1$. To quantify different modes of droplet dynamics, we introduce three order parameters. The first two, ψ_χ and $\psi_{\delta\phi}$, characterize the spatiotemporal order of the agents' rotational axes and their relative phases, respectively:

$$\psi_\chi = \left\langle \cos^2 \left(\frac{1}{N} \sum_{i=1}^N \chi_i(t) \right) \right\rangle_t \quad \text{and} \quad \psi_{\delta\phi} = \left\langle \cos^2 \left(\frac{1}{2(N-1)} \sum_{i=1}^{N-1} \delta\phi_{i,i+1}(t) \right) \right\rangle_t, \quad (5)$$

where $\langle \cdot \rangle_t$ denotes a time average. Apparently, ψ_χ describes how well the rotational axes $\mathbf{m}_i$ align with the z -axis, and $\psi_{\delta\phi}$ characterizes the phase differences between neighboring agents.

A third order parameter, ψ_σ , is introduced to quantify the temporal stability of the phase relationship. ψ_σ characterizes the dynamic fluctuations of the relative phase $\delta\phi_{i,i+1}(t)$ in the time domain. To distinguish between stable phase-locking (minimal fluctuations) and disorder dynamics (maximal fluctuations), we define the order parameter using the normalized standard deviation:

$$\psi_\sigma = 1 - \frac{\bar{\sigma}}{\sigma_{\max}}. \quad (6)$$

Here, $\bar{\sigma} = \frac{1}{N-1} \sum_{i=1}^{N-1} \sigma_i$ represents the averaged standard deviation of the phase differences within a single experiment, where $\sigma_i = \sqrt{\left\langle (\delta\phi_{i,i+1}(t) - \langle \delta\phi_{i,i+1} \rangle_t)^2 \right\rangle_t}$ denotes the standard deviation of the phase difference for the i -th pair. The normalization factor $\sigma_{\max}$ corresponds to the maximum standard deviation observed across the range of confinement strengths K for a given number of droplets N .

For systems with $l = 2.5R$ ($\forall i \in \{1, 2, \dots, N-1\}$), spring constants K ranging from 1 to 100, and droplet numbers N from 2 to 10, we find four dynamic modes of the active agents, which are ranked in below from most ordered to least ordered: synchronization (S), phase-locking (P), reorientation (R), and disordered (D) mode. In the following we elaborate these modes.

Synchronization (S) mode: The system achieves full synchronization as evidenced by the order parameters ψ_χ , $\psi_{\delta\phi}$ and ψ_σ asymptotically approaching unity (Fig. 3(a), $N = 3$ case), which correspond to a complete alignment of the rotational axes of all droplets along the z -axis, neighboring agent phases approach a near-zero lag, and remain temporally stable. This emergent coordination between the droplets originates from their hydrodynamic interactions that are similar to the wall-induced reorientation mechanism described in Section 2.1. To bridge the two scenarios, we can regard the wall as an infinitely large droplet; therefore, the alignment turns the active droplet to rotate along the line connecting the two droplet centers. (Supplementary Movie S4)

Phase-locking (P) mode: While the rotational axes of the droplets remain aligned with the z -axis, as in the synchronized mode, the phase-difference order parameter $\psi_{\delta\phi}$ is stabilized at intermediate values between zero and unity (Fig. 3(d) and Supplementary Movie S5).

Reorientation (R) mode: At elevated spring constant K and droplet number N , the system enters a dynamically frustrated regime where axial alignment persists while phase coordination becomes fundamentally compromised. Although droplets maintain collective orientation (Fig. 3(a) and Supplementary Movie S6), their phase relationship exhibits persistent oscillations ($\psi_{\delta\phi} \in [0, 1]$) accompanied by enhanced phase dominant fluctuation frequency (ψ_σ decreasing in Fig. 3(d) and Fig. S7(i)).

Disordered (D) mode: Emergent structural disorder dominates in large N and large K where neither synchronization nor phase-locking criteria are satisfied (Supplementary Movie S7), which is characterized by rotational axis misalignment ($\psi_\chi < 1$) and temporal fluctuations of the phase ($\psi_{\delta\phi} \in [0, 1]$ with minimized ψ_σ values as shown in

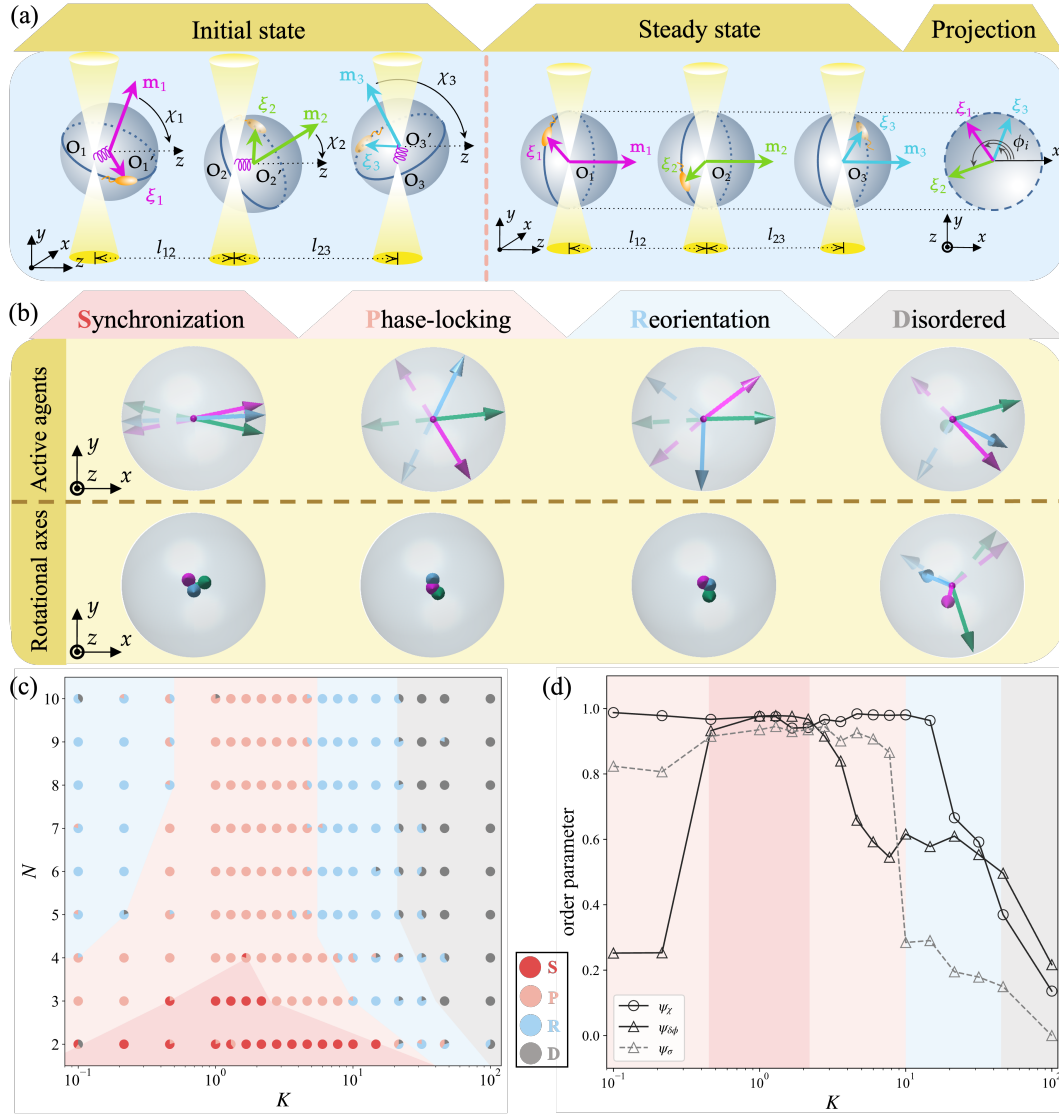


Fig. 3 (a) (Left panel) Three active droplets ($N = 3$) are confined by optical tweezers positioned uniformly along the x -axis, each with initially random pitched angles χ_i and random active agent positions ξ_i . (Right panel) An illustration depicting droplets reaching a potential stable mode (S, P, and R modes), each characterized by a phase ϕ_i and a phase difference $\delta\phi_{i,i+1} = \phi_{i+1} - \phi_i$. (b) Active agent vectors and rotational axis vectors (collapsed onto a unit sphere) for the S, P, R, and D modes. Dashed/faded arrows indicate vector orientations at half-period intervals preceding the current configuration, represented by solid arrows (Supplementary Information Section 9). (c) Steady state phase diagram for active droplets array, as a function of the confinement coefficient K and the number of droplets N . The spheres are color-coded to represent different modes: red for S, pink for P, blue for R, and gray for D. Spheres with two colors, such as pink and blue, indicate regions where both corresponding modes were observed. (d) Order parameters ψ_χ , $\psi_{\delta\phi}$, and ψ_σ for the $N = 3$ case, illustrating the sequential mode transitions P-S-P-R-D.

Fig. 3(d)).

In below we discuss more details of these dynamic modes. The simplest case, $N = 2$, is analogous to the droplet-wall case discussed in Section 2.1. The rotational axes of the two droplets tend to align along the line connecting their COMs at an intermediate K —This resembles the fact that an intermediate K can facilitate the droplet to rotate with respect to the wall normal direction. This resemblance can be understood by regarding the wall as another droplet of infinite radius. An extremely small or large K will destroy such alignment mode. A similar trend is also found in other N values, as the ordered modes S and P are only observed at intermediate K 's. These results indicate that constraining springs play an important role in the synchronization and alignment among droplets.

Another control parameter of the dynamic modes is the droplet number N . We find that a larger N will give rise to a less ordered mode. In fact, the alignment effect is due to near-field hydrodynamic effects. For $N > 2$, the next-or non-neighboring droplets tend to suppress the alignment effect, leading to less ordered dynamic modes.

Detailed analysis of the steady state reveals that there is a phase difference between the active agent and the droplet's orbital motion, i.e., $\langle \phi_{i,\text{agent}} \neq \phi_{i,\text{COM}} \rangle$ (Fig. 4(a,b)). This counter-clockwise phase shift originates from the force balance required to sustain the orbital dynamics. The active propulsion force $\mathbf{F}_{\text{act}}$, generated by surface-active flows induced by the active agent, must simultaneously counterbalance the radial restoring force $\mathbf{F}_{\text{trap}}$ from the optical trap and the tangential hydrodynamic drag force, $\mathbf{F}_{\text{drag}}$. Consequently, the interplay between this dynamical force balance and the agent's geometric orientation dictates a characteristic phase lead relative to the droplet's center of mass. This analysis aligns well with the simulation results, as illustrated by the dashed lines in Fig. 4(b)

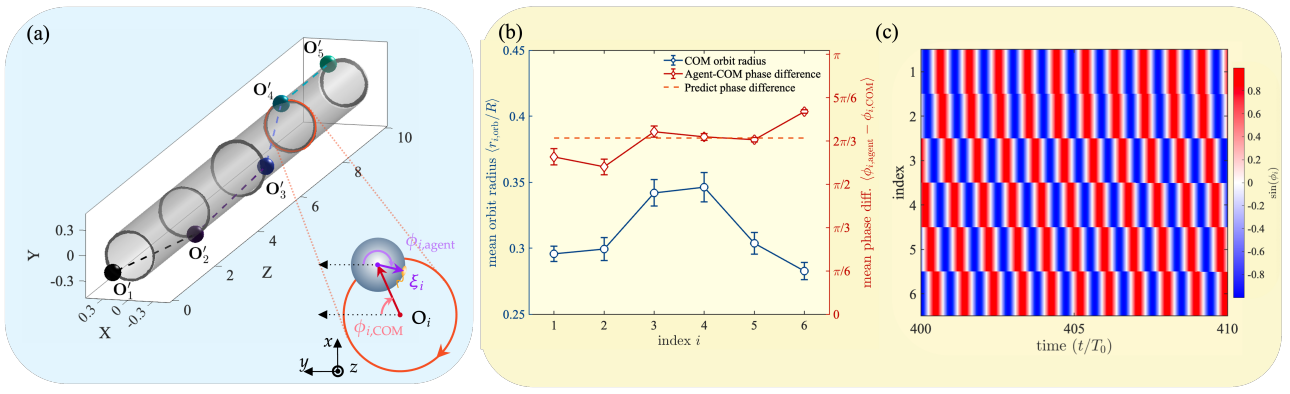


Fig. 4 (a) Snapshot of the collective motion for an array of $N = 5$ droplets in the phase-locking mode. The centers of mass (COM) of the droplets (O'_i , marked by spheres) follow circular trajectories (solid gray lines) that collectively define a cylindrical surface (light gray shading). The auxiliary line connecting adjacent COMs (dashed line) reveals an emergent helical geometry. The inset (not to scale) defines the coordinate system: $\phi_{i,\text{agent}}$ represents the phase of the active agent relative to the droplet center, and $\phi_{i,\text{COM}}$ denotes the orbital phase of the droplet COM relative to the optical trap center O_i . (b) Spatial distribution of the dynamics for an $N = 6$ system. The blue curve (left axis) shows the mean orbital radius $r_{i,\text{orb}}/R$ for each droplet. The red curve (right axis) depicts the mean phase difference between the active agent and the droplet's orbital motion, $\langle \phi_{i,\text{agent}} - \phi_{i,\text{COM}} \rangle$. The dashed lines represent the theoretical predictions derived from the force balance between the active propulsion, trap restoring force, and hydrodynamic drag. (c) Spatiotemporal evolution of the phase $\sin(\phi_i(t))$ for an $N = 6$ array. The diagonal stripe pattern indicates a constant phase shift between neighbors, characteristic of a metachronal wave formation.

(see Supplementary Information Section 4.4 for detailed derivation). This signifies persistent, non-vanishing phase differences between droplet pairs—a hydrodynamic analogue to traveling waves with fixed wavelengths, biologically recognized as metachronal waves [38, 41] (Fig. 4(c)). The spatiotemporal evolution of phases in Fig. 4(c) exhibits characteristic diagonal stripes, confirming the formation of a stable wave propagating through the droplet array.

A distinct feature differentiating the synchronized and phase-locked modes lies in the collective motion of the droplets' COMs. In the synchronized mode, the COMs of all droplets remain in phase. Due to finite-size boundary effects, the orbital radii vary across the array: smaller at the ends and larger in the middle (Fig. 4(b)), causing the collective trajectory envelope to form a spindle-like surface rather than a perfect cylinder. The phase-locked mode exhibits a dynamic helical geometry: at each timestep, the COMs are organized on a helical-like line, reflecting both spatial periodicity and progressive phase offsets between neighboring droplets (Fig. 4(a)). We observe that the chirality of this emergent helix is not predetermined; rather, it is spontaneously selected based on random initial conditions and shows no statistical bias across ensembles (Supplementary Information Section 11). This spatial chirality is intrinsically coupled to the temporal order of the system. The sign of the inter-droplet phase difference determines both the direction of the metachronal wave propagation (indicated by the slope direction in Fig. 4(c)) and the handedness of the helical structure formed by the instantaneous COM positions. Thus, the emergence of a specific helical chirality serves as a geometric signature of the wave's propagation direction.

These emergent geometries—cylindrical symmetry in the synchronization mode versus helical order in phase locking mode—encode intrinsic hydrodynamic constraints on the collective dynamics. The helical arrangement, in particular, suggests a natural mechanism for propagating phase perturbations through the array, a concept leveraged in later spatial control strategies (Section 2.3). Such structural differences further underscore how dissipation minimization governs not only temporal coordination but also spatial organization in active droplet systems.

To quantitatively elucidate the physical origin of these modes, we develop a reduced theory based on cycle-averaged hydrodynamic interactions (Supplementary Information Section 6). The theory separates the collective dynamics into two hierarchical processes. The first is alignment of the rotational axes. The trap-constrained orbital motion modulates the pair mobility between neighboring droplets; after averaging over one orbit, this produces dissipative alignment couplings ($J_{\parallel}(K, l)$) and ($J_m(K, l)$). Here $J_{\parallel}$ aligns each rotational axis with the line of centers, whereas J_m aligns neighboring axes with each other. Both couplings inherit the same stiffness dependence,

$$J_{\alpha}(K, l) = J_{\alpha,0}(l) \frac{K}{K^2 + (\gamma_t \Omega_0)^2}, \quad \alpha \in \{\parallel, m\}. \quad (7)$$

This form follows from the scaling $J_{\alpha} \propto K r_{\text{orb}}^2(K)$.

The second process is locking of the active-agent phases. For the phase difference $\Delta = \phi_{i+1} - \phi_i$, the leading phase-reduced dynamics is

$$\dot{\Delta} = \Delta \Omega_{\text{eff}} - \Gamma_1(K, l) \sin \Delta - \Gamma_2(K, l) \sin(2\Delta). \quad (8)$$

Here Γ_1 is the leading phase-attractive coupling generated by the same trap-mediated rectification mechanism, while Γ_2 represents the leading finite-size and active-slip correction. The drift term $\Delta \Omega_{\text{eff}}$ represents differential precession and many-body frustration. For an ideal deterministic pair of identical droplets, $\Delta \Omega_{\text{eff}} = 0$; in arrays, non-neighbor interactions generate a finite effective local drift.

We therefore define two ordering ratios. The first,

$$\mathcal{A}(K, l) = \frac{J_{\parallel}(K, l)}{J_{\chi, c}}, \quad (9)$$

controls rotational-axis alignment and corresponds to the simulation order parameter ψ_{χ} . The second,

$$\mathcal{R}(K, l) = \frac{\Gamma_1(K, l)}{|\Delta\Omega_{\text{eff}}|}, \quad (10)$$

controls active-agent phase locking. Because both $J_{\parallel}$ and Γ_1 originate from $Kr_{\text{orb}}^2(K)$, both ratios have the same non-monotonic stiffness dependence,

$$\mathcal{A}(K, l) \sim \mathcal{A}_{\text{max}}(l) \frac{2K\gamma_t\Omega_0}{K^2 + (\gamma_t\Omega_0)^2}, \quad \mathcal{R}(K, l) \sim \mathcal{R}_{\text{max}}(l) \frac{2K\gamma_t\Omega_0}{K^2 + (\gamma_t\Omega_0)^2}, \quad (11)$$

and both are maximized near $K_{\text{opt}} \simeq \gamma_t\Omega_0$.

This two-level criterion explains the observed mode sequence. When $\mathcal{A} < 1$, rotational axes are not robustly aligned and the system enters the D mode. When $\mathcal{A} > 1$ but $\mathcal{R} < 1$, the axes remain aligned but the active-agent phases drift or oscillate, corresponding to the R mode. When $\mathcal{A} > 1$ and $\mathcal{R} > 1$, the active-agent phase difference admits a stable fixed point. In the first-harmonic approximation, the stable phase lag satisfies $|\sin \Delta_{\text{eq}}| = \frac{1}{\mathcal{R}}$. We therefore introduce a small phase tolerance Δ_S to distinguish near-zero-lag synchronization from finite-lag phase locking. The corresponding synchronization threshold is $\mathcal{R}_S = \frac{1}{\sin \Delta_S}$. Thus, $1 < \mathcal{R} < \mathcal{R}_S$ gives a stable finite-lag phase-locked state, corresponding to the P mode, whereas $\mathcal{R} > \mathcal{R}_S$ gives a near-zero-lag synchronized state, corresponding to the S mode. As K is increased at fixed N , the non-monotonic form of equation (11) naturally produces ordered states only within an intermediate stiffness window. As N increases, many-body frustration reduces the effective values of $\mathcal{A}_{\text{max}}$ and $\mathcal{R}_{\text{max}}$, making the system more likely to enter R or D modes. The dynamic diagram presented in Fig. 3(c) summarizes the mode distribution for varying numbers of droplets in an array, with the confinement coefficient K ranging from 1 to 100. In general, when the number of droplets in the array is controlled, the system undergoes a phase transition sequence of D-R-P-S-P-R-D as the confinement strength increases from weak to strong. As the confinement strength on each droplet is controlled and the number of droplets in the array is reduced, the system is more likely to reach the S or P mode.

2.3 Regulating phase transition with spatial perturbation

Building upon the emergent helical geometry intrinsic to the phase-locked mode (Section 2.2), we develop a hydrodynamic control strategy through targeted spatial perturbations. The spontaneous helical coordination—where droplet COMs form circular helices with progressive phase offsets—provides a geometric template for steering collective dynamics. We introduce two perturbation strategies: helical perturbation (Fig. 5(a)) and axial perturbation (Fig. 5(b)).

First, for the helical perturbation, we actively tune the system's dissipation landscape by parametrically modulating the helix dimensions through radius R_d and pitch $P_d = N_d l_{i, i+1}$ (Fig. 5(a)). Increasing R_d reduces helical curvature, which reorients hydrodynamic coupling from predominantly axial (z -direction) to lateral (xy -plane) components. This curvature-mediated flow restructuring modifies the effective interaction symmetry. Simultaneously, the helix pitch P_d regulates positional matching between adjacent droplets in the xy -plane, creating staggered flow fields that stabilize non-zero phase differences.

Second, we propose an axial perturbation strategy (Fig. 5(b)), where the uniformity of the trapping centers' displacement is altered along the z -axis. In the unperturbed configuration, trapping centers are uniformly distributed along the z -axis with spacing $l_{i, i+1} = 2.5R$. For the axial perturbation, we maintain fixed positions for the trapping centers at both ends of the droplet array while adjusting the internal trap centers to reduce the inter-droplet distance l_{ij} for specific pairs, thereby creating a dimerized chain structure. This axial clustering alters the near-field hydrodynamic coupling strength between neighbors.

Simulation results demonstrate that these spatial perturbation strategies can effectively regulate phase transitions, as shown in the time-evolution of order parameters in Fig. 5(c-e):

- (i) Phase-locking to synchronization (P→S): This transition is achieved exclusively via helical perturbation (Fig. 5(c) and Supplementary Movie S8). The reduced curvature (large R_d) weakens lateral flow coupling, allowing near-field interactions to enforce phase alignment. The axial perturbation fails to induce this transition, as evidenced by the high residual phase difference (Fig. 5(f)).
- (ii) Reorientation to phase-locking (R→P): Both helical and axial perturbations successfully stabilize the system from the frustrated reorientation mode into a phase-locked mode (Fig. 5(d) and Supplementary Movie S9). For the axial case, the formation of interaction pairs strengthens local coupling, suppressing the persistent oscillations of the rotational axes.

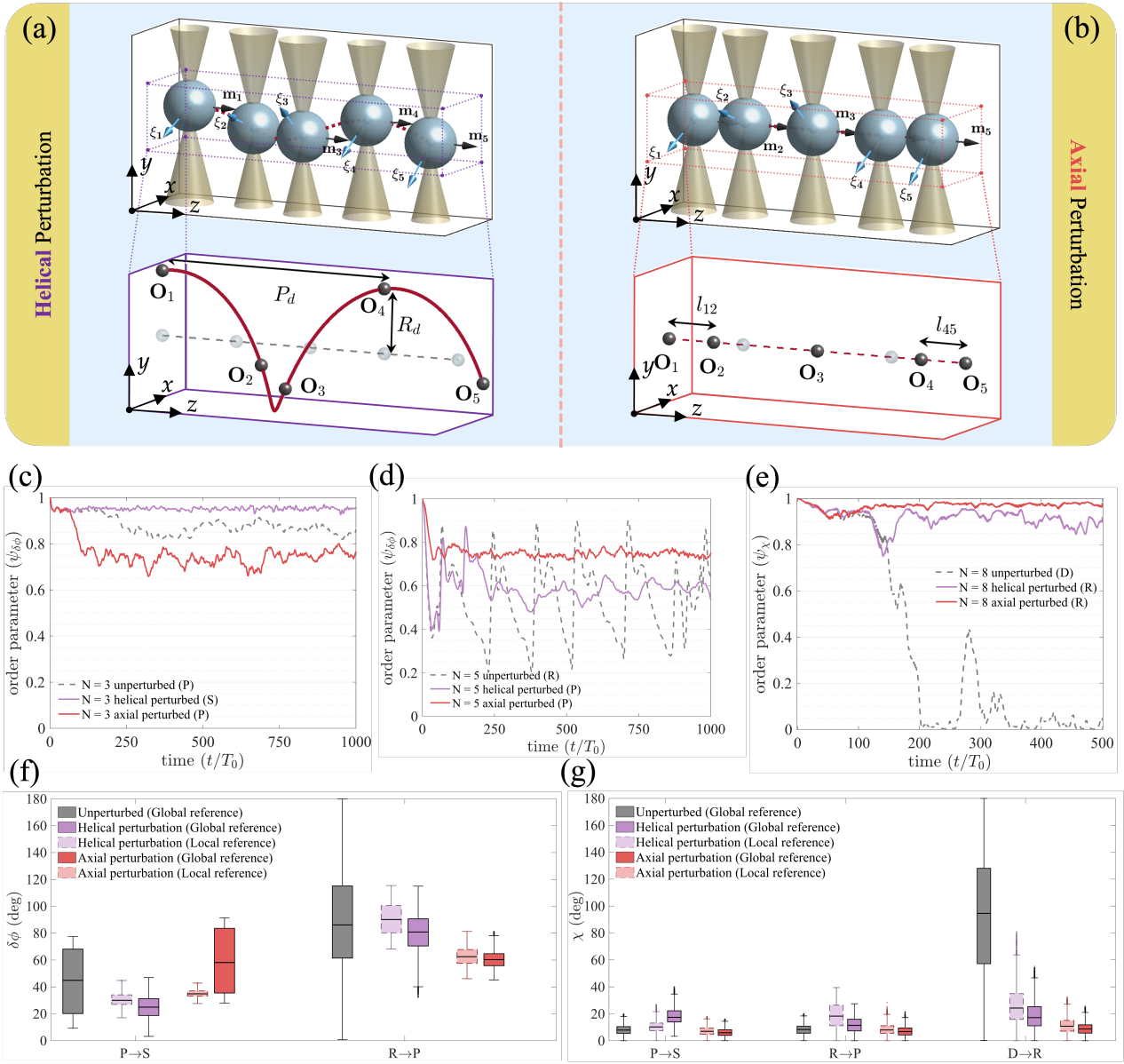


Fig. 5 (a) Configuration of helical perturbation ($N = 5$): Traps follow a circular helix with pitch P_d and radius R_d . Parameter definitions are schematized in the bottom panel (schematic representation). (b) Configuration of axial perturbation ($N = 5$): Internal trap centers are shifted to form interacting pairs (adjusting spacing l_{ij}) while keeping end droplets fixed. Parameter definitions are schematized in the bottom panel (schematic representation). (c-e) Time evolution of order parameters showing mode transitions. Unperturbed cases are dashed lines; perturbed cases are solid lines. (c) P→S transition ($N = 3$) characterized by $\psi_{\delta\phi}$. Helical perturbation ($P_d = 3, R_d = 0.03R$) achieves synchronization (red solid), while axial perturbation ($l_{12} = 2.45R$) remains phase-locked (pink solid). (d) R→P transition ($N = 5$) characterized by $\psi_{\delta\phi}$. Both helical ($P_d = 3, R_d = 0.10R$) and axial ($l_{12} = l_{45} = 2.3R$) perturbations drive the system from reorientation mode to the phase-locking mode. (e) D→R transition ($N = 8$) characterized by ψ_{χ} . Both helical ($P_d = 6, R_d = 0.02R$) and axial ($l_{12} = l_{34} = l_{56} = 2.4R$) perturbations drive the system from disorder mode to the reorientation mode. (f-g) Statistical box plots of steady-state dynamics comparing global (lab-frame) and local (neighbor-referenced) metrics. The box represents the interquartile range (25th to 75th percentile), the horizontal line indicates the median, and whiskers extend to the data range. (f) Distribution of phase difference $\delta\phi$. Global reference corresponds to the definition $\delta\phi_{i,i+1} = \phi_{i+1} - \phi_i$ used in order parameters, while the local reference is defined as the angle between the active agent and the neighbor's agent vector projected onto the instantaneous rotational plane. (g) Distribution of pitch angle χ of the rotational axis. Global reference denotes the angle between the rotational axis and the z -axis, while the local reference denotes the angle between neighboring rotational axes.

354(iii) Disordered to Reorientation (D→R): Both strategies guide the system toward a more ordered reorientation (R)
 355 mode (Fig. 5(e) and Supplementary Movie S10).

356 To elucidate the mechanism driving these transitions, we analyze the statistical distributions of the active agent
 357 phase difference $\delta\phi$ and the rotational axis pitch angle χ in the steady state, comparing global lab-frame references
 358 with local neighbor-based references (Fig. 5(f, g)). For the phase difference $\delta\phi$, the global reference corresponds to
 359 the standard phase difference $\delta\phi_{i,i+1} = \phi_{i+1} - \phi_i$ as defined in the order parameter, whereas the local reference
 360 quantifies the alignment of an agent relative to its neighbor's agent on the neighbor's instantaneous rotational plane.
 361 For the pitch angle χ , the global reference measures the alignment of the rotational axis with the z -axis, while the
 362 local reference measures the angle between adjacent rotational axes.

363 As shown in Fig. 5(f), successful transitions (e.g., P→S via helical perturbation and R→P via both helical and
 364 axial perturbations) are characterized by narrowing of the $\delta\phi$ distribution. This signifies an enhancement in local

orderliness, indicating that spatial perturbations effectively suppress fluctuations of the phase difference between neighboring droplets' active agents. This local stabilization facilitates the propagation of coherence across the array, resulting in a pronounced improvement in global orderliness, as evidenced by the decrease and stabilization of the global phase differences (Fig. 5(f)). These results imply that by regularizing short-range hydrodynamic interactions, the perturbations confine the system to the basin of attraction of the globally ordered mode. In the D→R transition (Fig. 5(g)), both perturbation methods reduce the variance of χ compared to the unperturbed case, restoring orientational order. We observe a geometric trade-off in the helical P→S and R→P transitions: while the helical perturbation enables global synchronization or phase-locking, the imposed helical geometry induces a slight increase in rotational axes' local misalignment compared to the unperturbed case (Fig. 5(g)). This suggests that the control strategy works not by enforcing perfect local alignment of rotational axes, but by imposing a robust geometric template that reduces the phase differences, thereby facilitating the emergence of global order.

The observed transition reversibility (Supplementary Information Section 13) confirms the non-destructive nature of this geometric control scheme. While helical perturbation offers a versatile control over chirality and lateral coupling, axial perturbation provides a simpler, one-dimensional mechanism to enhance local ordering and stabilize collective modes.

3 Methods

In the numerical simulations, droplets are modeled as rigid spheres, with an active neutral swimmer (mimicking microswimmers such as a Paramecium [42]) attached to their surfaces. Physically, this stationary attachment can be naturally realized through the quasi-two-dimensional membrane hydrodynamics of the interface [43]. Assuming a three-layer fluid system, the active agents reside within a thin interfacial layer of 2D viscosity η_m^{2D} , sandwiched between a highly viscous droplet interior (η_{in}) and an external low-viscosity bulk fluid (η_{out} , e.g., water). According to fluid membrane theory, the translational motility of an interfacial swimmer is regulated by the Saffman length $l_S = \eta_m^{2D}/(\eta_{in} + \eta_{out})$. In the practical limit of an extremely viscous droplet core ($\eta_{in} \gg \eta_{out}$), the immense viscous traction from the internal phase severely suppresses the Saffman length, entering the regime $a/l_S \gg 1$ (where a is the characteristic radius of the agent). In this regime, the translational velocity of the agent relative to the interface is damped, stalling its lateral displacement. The localized active strokes of the agents continue to drive the surrounding low-viscosity interfacial fluid. This enables the agents to function as stationary hydrodynamic pumps that sustain the prescribed active slip flow on the spherical interface without translating along it. Persistent interfacial localization can also be achieved through biochemical adhesion. In bacteria-covered oil droplets, cell hydrophobicity and oil-water interfacial tension control bacterial attachment to the interface, allowing the adhered population to transmit growth-generated active stresses to the droplet and drive large-scale deformation [25]. In the present minimal model, the microscopic anchoring mechanism is coarse-grained: the active agent is held fixed relative to the droplet while its actuation is represented by a prescribed tangential slip flow.

The spheres are immersed in a Stokes flow (characterized by $Re \ll 1$). For single-sphere configurations, the sphere is initially suspended at height h_0 above a no-slip bottom wall. For multi-sphere systems, we consider free-space scenarios where only inter-body hydrodynamic interactions are present. Instantaneous velocities (in static simulations) and dynamic trajectories (in dynamic simulations) are determined by solving a mobility problem using the Rigid Multiblob method [27, 28], which we customize to incorporate restoring forces and slip boundary conditions.

For unbounded domain, the droplets are immersed in infinitely large domain of Stokes flow. The fluid dynamics is governed by the Stokes equations for low Reynolds number [44, 45]:

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}, \quad \nabla \cdot \mathbf{u} = 0, \quad (12)$$

where the stress tensor $\boldsymbol{\sigma} = -p\mathbf{I} + \eta[\nabla\mathbf{u} + (\nabla\mathbf{u})^\top]$ incorporates the pressure field p and dynamic viscosity η . The flow on the spherical droplet surface is generated by the active microswimmer and is regarded as a purely surface flow. Importantly, this surface flow remains unperturbed by bulk flow or droplet movement. To establish the surface flow field induced by active agents, we first consider the governing equations for an incompressible surface flow on a unit sphere. Let $\boldsymbol{\nu}(\theta, \phi)$ denote the tangential velocity field on the spherical surface, which satisfies the surface divergence-free condition:

$$\nabla_S \cdot \boldsymbol{\nu} = 0, \quad (13)$$

where ∇_S is the surface gradient operator. For Stokes flow, the tangential stress balance reduces to a fourth-order biharmonic equation for the stream function $\Psi(\theta, \phi)$ [46]:

$$\Delta_S^2 \Psi = 0, \quad (14)$$

where Δ_S is the surface Laplacian. Equation (14) admits solutions expressed in terms of spherical harmonics $Y_l^m(\theta, \phi)$:

$$\Psi(\theta, \phi) = \sum_{l=1}^{\infty} \sum_{m=-l}^l \left(A_{lm} r^l + B_{lm} r^{-(l+1)} \right) Y_l^m(\theta, \phi). \quad (15)$$

For flows confined to the unit sphere ($r = 1$), the coefficients simplify to $\Psi(\theta, \phi) = \sum_{l=1}^{\infty} \sum_{m=-l}^l C_{lm} Y_l^m(\theta, \phi)$. For the specific problem of a flow driven by a point-like active agent, it is more direct to solve for the flow field using a

Green's function approach. We model the active agent as a source dipole on the sphere's surface [47]. A fundamental component of this model is a source monopole, which corresponds to a Dirac delta function singularity in the surface pressure field. The velocity field can then be derived from the gradient of this pressure potential. For a monopole source at (θ_0, ϕ_0) , the associated Green's function $G(\theta, \phi; \theta_0, \phi_0)$ satisfies:

$$\Delta_S G = -\delta(\cos \theta - \cos \theta_0) \delta(\phi - \phi_0), \quad (16)$$

where the negative sign ensures consistency with conventional Green's function definitions. Expanding in spherical harmonics and using their completeness relation:

$$\delta(\cos \theta - \cos \theta_0) \delta(\phi - \phi_0) = \sum_{l=0}^{\infty} \sum_{m=-l}^l Y_l^m(\theta_0, \phi_0) Y_l^{m*}(\theta, \phi), \quad (17)$$

we obtain the spectral solution:

$$G(\theta, \phi; \theta_0, \phi_0) = -\frac{1}{4\pi} \sum_{l=1}^{\infty} \frac{2l+1}{l(l+1)} P_l(\cos \gamma), \quad (18)$$

where P_l are Legendre polynomials and γ is the angular distance between (θ, ϕ) and (θ_0, ϕ_0) . The $l = 0$ mode is excluded to enforce the divergence-free condition. Equation (18) can be summed explicitly to yield the closed-form expression [48, 49]:

$$G(\theta, \phi; \theta_0, \phi_0) = \frac{1}{4\pi} \log(1 - \cos \gamma) + \text{constant}, \quad (19)$$

The velocity field associated with a source monopole derives from the surface gradient of G :

$$\boldsymbol{\nu}_{\text{monopole}} = \nabla_S G = \frac{\partial G}{\partial \gamma} \nabla_S \gamma, \quad (20)$$

explicitly given by:

$$\boldsymbol{\nu}_{\text{monopole}} = -\frac{1}{4\pi} \frac{\sin \gamma}{1 - \cos \gamma} \hat{\gamma}, \quad (21)$$

where $\hat{\gamma}$ is the azimuthal unit vector perpendicular to the great circle connecting (θ, ϕ) and (θ_0, ϕ_0) . A source dipole field is then constructed by superposing two oppositely oriented monopoles of strength $\pm q$ separated by an infinitesimal distance δ_p along a chosen axis. Taking the limit $\delta_p \rightarrow 0$ while maintaining a finite dipole strength $v_s = q\delta_p$, we obtain:

$$\boldsymbol{\nu}_{\text{dipole}} = \lim_{\delta_p \rightarrow 0} [\boldsymbol{\nu}_{\text{monopole}}(\mathbf{r} + \delta_p/2) - \boldsymbol{\nu}_{\text{monopole}}(\mathbf{r} - \delta_p/2)] = v_s \frac{\partial}{\partial \delta_p} \boldsymbol{\nu}_{\text{monopole}}. \quad (22)$$

The flow field is numerically discretized and imposed as slip boundary conditions on the rigid spheres' surfaces. To ensure the fidelity of our numerical scheme, we validated the implementation of the source dipole slip boundary condition by comparing the simulated propulsion velocities with theoretical predictions derived from the reciprocal theorem [31]. Each droplet i carries a prescribed tangential slip velocity $\boldsymbol{\nu}_i(\mathbf{r})$, generated by an active agent located at $\boldsymbol{\xi}_i$ and defined relative to the droplet surface. In the laboratory frame, this active slip is superimposed on the rigid-body translation and rotation of the droplet. The boundary condition on the droplet surface S_i is therefore [27]

$$\mathbf{u}(\mathbf{r}) = \mathbf{U}_i + \boldsymbol{\Omega}_i \times (\mathbf{r} - \mathbf{x}_i) + \boldsymbol{\nu}_i(\mathbf{r}), \quad \mathbf{r} \in S_i, \quad (23)$$

where $\mathbf{U}_i$ and $\boldsymbol{\Omega}_i$ denote the translational and angular velocities of droplet i , respectively. For systems with a stationary bottom wall, the no-slip condition $\mathbf{u} = \mathbf{0}$ is imposed at $z = 0$. The rigid-body velocities are determined together with the surface tractions by enforcing the instantaneous force and torque balances

$$\int_{S_i} \boldsymbol{\sigma} \cdot \mathbf{n} dS = \mathbf{F}_{\text{ext},i}, \quad \int_{S_i} (\mathbf{r} - \mathbf{x}_i) \times (\boldsymbol{\sigma} \cdot \mathbf{n}) dS = \mathbf{T}_{\text{ext},i}, \quad (24)$$

where $\mathbf{F}_{\text{ext},i}$ and $\mathbf{T}_{\text{ext},i}$ are the net externally applied force and torque on droplet i , and $\mathbf{n}$ is the outward surface normal. For an untethered droplet with no externally applied force or torque, $\mathbf{F}_{\text{ext},i} = \mathbf{T}_{\text{ext},i} = \mathbf{0}$, and the resulting motion is both force-free and torque-free. The presence of a stationary no-slip wall modifies the hydrodynamic mobility but does not, by itself, alter these force- and torque-free conditions. For an optically trapped droplet, $\mathbf{F}_{\text{ext},i} = \mathbf{F}_{\text{trap},i}$ is generally nonzero, so the droplet is not force-free; instead, the hydrodynamic drag force instantaneously balances the restoring force. Because the restoring force is applied through the droplet center and no external couple is imposed, $\mathbf{T}_{\text{ext},i} = \mathbf{0}$, and the trapped droplet remains torque-free. For a freely swimming sphere in an unbounded fluid, the force- and torque-free conditions can also be understood through the pumping-swimming decomposition of the squirmer flow [32]. The Stokeslet (translational) and rotlet (rotational) contributions present in the fixed-body pumping problem are cancelled by the flows associated with the induced rigid-body translation and rotation,

respectively. This cancellation removes the flow singularities associated with a net force and a net torque, while force- and torque-free higher-order contributions, including the residual source-dipole flow, remain in the swimming disturbance field. Owing to the linearity of the Stokes equations, the externally applied forces and torques and prescribed surface slip are related affinely to the rigid-body velocities through the mobility problem

$$\begin{bmatrix} \mathbf{U}_i \\ \boldsymbol{\Omega}_i \end{bmatrix} = \mathcal{M}_i \begin{bmatrix} \mathbf{F}_{\text{ext},i} \\ \mathbf{T}_{\text{ext},i} \end{bmatrix} - \widetilde{\mathcal{M}}_i[\boldsymbol{\nu}_i]. \quad (25)$$

Here, $\mathcal{M}_i$ maps the externally applied force and torque to the rigid-body motion of droplet i , whereas $\widetilde{\mathcal{M}}_i$ gives the rigid-body response induced by the prescribed surface slip under the corresponding force- and torque-balance constraints. By solving this mobility problem, we determine the instantaneous translational and angular velocities of each droplet. The droplet's trajectory and orientation are then updated kinematically. The position of the droplet's center of mass, $\mathbf{x}_i$, evolves according to its linear velocity $\mathbf{U}_i$. Simultaneously, any orientation vector $\mathbf{p}_i$ fixed to the body of the droplet, such as the rotational axis $\mathbf{m}_i$, rotates with the angular velocity $\boldsymbol{\Omega}_i$. The complete kinematic update is thus given by:

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{U}_i \quad \text{and} \quad \frac{d\mathbf{p}_i}{dt} = \boldsymbol{\Omega}_i \times \mathbf{p}_i, \quad (26)$$

closing the system of equations that simultaneously describe the fluid flow, active slip dynamics, and droplet motions.

4 Discussion

This study provides a comprehensive examination of the hydrodynamic interactions of droplets laden with active agents, with a focus on understanding the roles of both artificial and biological materials and the intriguing phenomena of synchronization/phase-locking. We developed and presented a detailed model setup and simulation methods specifically for droplets with surface active agents. Our simulation results not only qualitatively align with existing experimental findings but also reveal complex dynamic behaviors in both unbounded and semi-bounded Stokes flows, emphasizing the intricate nature of these interactions.

We have demonstrated that hydrodynamic interactions can lead to diverse outcomes such as synchronization, phase locking, reorientation, and disordered mode among multiple active droplets arranged in an array. These findings suggest that the behavior of active droplets is not only rich but also highly sensitive to initial conditions and interactions, opening new avenues for controlling their dynamics. Similar to recent findings where quenched disorder induces synchronized rotations in dense active matter [12], we propose employing spatial perturbations as a novel method to control phase transitions, potentially offering a mechanism to guide the behavior of active droplets for specific applications.

Our research sheds light on the collective dynamics and interaction behaviors of droplets driven by surface-anchored active agents, offering valuable insights for both fundamental understanding and practical applications. Future research will focus on exploring the underlying mechanisms that drive reorientation and synchronization phenomena, providing a deeper understanding of the processes involved. Additionally, we plan to modify our experimental setup from a uniformly distributed linear array to an $N \times N$ carpet array, which will allow us to explore more complex arrangements. We will also investigate scenarios where multiple active agents are anchored, periodically moving, or coupled with bulk fluid motion on the droplet surface, aiming to uncover the resultant dynamics and further elucidate the nature of hydrodynamic interactions in these systems.

Supplementary information. Please see active_droplet_SI.tex

Supplementary movies. Please see directory SI_movies

Single droplet dynamics

- **Movie S1** (Fig. 1(b)): *Circular trajectories in an unbounded domain*
SI_movie_1_single_active_droplet_unconfined_unbounded.mp4
- **Movie S2** (Fig. 1(c)): *Petal-like trajectories under wall confinement*
SI_movie_2_single_active_droplet_unconfined_bounded_chi0_045_h0_01d10.mp4
Petal-shaped trajectory with initial configuration pitch angle $\chi_0 = \pi/4$ and $\bar{h}_0 = 1.10$.
- **Movie S3** (Fig. 2(b)): *Spontaneous reorientation under trap confinement*
SI_movie_3_single_active_droplet_confined_bounded_K_002d154_chi0_060_h_01d60
Reorientation toward wall normal (z -axis) under restoring force $K = 2.154$, trap center height $\bar{h}_0 = 1.60$ and initial pitch angle $\chi_0 = \pi/3$.

Multi-droplet array: steady states

- **Movie S4** (Fig. 3): *Synchronized state ($N=2$)*
SI_movie_4_S_mode_N2_K_002d154_uniform_lj_02d50.mp4
- **Movie S5** (Fig. 3): *Phase-locking state ($N=5$)*
SI_movie_5_P_mode_N5_K_002d154_uniform_lj_02d50.mp4

- **Movie S6** (Fig. 3): *Reorientation state* ($N=5$)
SI_movie_6_R_mode_N5_K_010d000_uniform_lij_02d50.mp4
- **Movie S7** (Fig. 3): *Disordered mode* ($N=8$)
SI_movie_7_D_mode_N8_K_031d622_uniform_lij_02d50.mp4

Regulating phase transition

- **Movie S8** (Fig. 5(c)): $P \rightarrow S$ transition
SI_movie_8_unperturbed_P_N3_K_002d783_uniform_lij_02d50.mp4
SI_movie_8_perturbed_PtoS_N3_K_002d783_helical_Rd_0d03_Pd_3.mp4
- **Movie S9** (Fig. 5(d)): $R \rightarrow P$ transition
SI_movie_9_unperturbed_R_N5_K_005d994_uniform_lij_02d50.mp4
SI_movie_9_perturbed_RtoP_N5_K_005d994_helical_Rd_0d10_Pd_3.mp4
- **Movie S10** (Fig. 5(e)): $D \rightarrow R$ transition
SI_movie_10_unperturbed_D_N8_K_031d622_uniform_lij_02d50.mp4
SI_movie_10_perturbed_DtoR_N8_K_031d622_helical_Rd_0d02_Pd_6.mp4

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Declarations

- Funding
- Conflict of interest/Competing interests (check journal-specific guidelines for which heading to use)
- Ethics approval and consent to participate
- Consent for publication
- Data availability
- Materials availability
- Code availability
- Author contribution

References

- [1] Vicsek, T., Zafeiris, A.: Collective motion. *Physics Reports* **517**(3-4), 71–140 (2012) <https://doi.org/10.1016/j.physrep.2012.03.004>
- [2] Wensink, H.H., Dunkel, J., Heidenreich, S., Drescher, K., Goldstein, R.E., Löwen, H., Yeomans, J.M.: Mesoscale turbulence in living fluids. *Proceedings of the National Academy of Sciences* **109**(36), 14308–14313 (2012) <https://doi.org/10.1073/pnas.1202032109>
- [3] Martín-Gómez, A., Levis, D., Díaz-Guilera, A., Pagonabarraga, I.: Collective motion of active brownian particles with polar alignment **14**(14), 2610–2618 <https://doi.org/10.1039/C8SM00020D> . Publisher: The Royal Society of Chemistry
- [4] Zhang, H.P., Be'er, A., Florin, E.-L., Swinney, H.L.: Collective motion and density fluctuations in bacterial colonies. *Proceedings of the National Academy of Sciences* **107**(31), 13626–13630 (2010) <https://doi.org/10.1073/pnas.1001651107> . Publisher: Proceedings of the National Academy of Sciences
- [5] Cavagna, A., Cimarelli, A., Giardina, I., Parisi, G., Santagati, R., Stefanini, F., Viale, M.: Scale-free correlations in starling flocks **107**(26), 11865–11870 <https://doi.org/10.1073/pnas.1005766107> . Publisher: Proceedings of the National Academy of Sciences
- [6] Lushi, E., Wioland, H., Goldstein, R.E.: Fluid flows created by swimming bacteria drive self-organization in confined suspensions **111**(27), 9733–9738 <https://doi.org/10.1073/pnas.1405698111>
- [7] Elfring, G.J., Lauga, E.: Hydrodynamic phase locking of swimming microorganisms **103**(8), 088101 <https://doi.org/10.1103/PhysRevLett.103.088101>
- [8] Uchida, N., Golestanian, R.: Synchronization and collective dynamics in a carpet of microfluidic rotors **104**(17), 178103 <https://doi.org/10.1103/PhysRevLett.104.178103>
- [9] Hydrodynamic synchronization at low reynolds number **7**(7), 3074 <https://doi.org/10.1039/c0sm01121e>
- [10] Chen, C., Liu, S., Shi, X.-q., Chaté, H., Wu, Y.: Weak synchronization and large-scale collective oscillation in dense bacterial suspensions **542**(7640), 210–214 <https://doi.org/10.1038/nature20817>

- [11] Samatas, S., Lintuvuori, J.: Hydrodynamic synchronization of chiral microswimmers **130**(2), 024001 <https://doi.org/10.1103/PhysRevLett.130.024001>
- [12] Vahabli, D., Vicsek, T.: Emergence of synchronised rotations in dense active matter with disorder **6**(1), 56 <https://doi.org/10.1038/s42005-023-01173-5>
- [13] Vilfan, A., Jülicher, F.: Hydrodynamic flow patterns and synchronization of beating cilia **96**(5), 058102 <https://doi.org/10.1103/PhysRevLett.96.058102>
- [14] Lauga, E., Powers, T.R.: The hydrodynamics of swimming microorganisms **72**(9), 096601 <https://doi.org/10.1088/0034-4885/72/9/096601>
- [15] Elgeti, J., Winkler, R.G., Gompper, G.: Physics of microswimmers—single particle motion and collective behavior: a review **78**(5), 056601 <https://doi.org/10.1088/0034-4885/78/5/056601> . Publisher: IOP Publishing
- [16] Hu, S., Meng, F.: Multiflagellate swimming controlled by hydrodynamic interactions
- [17] Matsui, H., Randell, S.H., Peretti, S.W., Davis, C.W., Boucher, R.C.: Coordinated clearance of periciliary liquid and mucus from airway surfaces. **102**(6), 1125–1131 <https://doi.org/10.1172/JCI2687>
- [18] Khelloufi, M.-K., Loiseau, E., Jaeger, M., Molinari, N., Chanez, P., Gras, D., Viallat, A.: Spatiotemporal organization of cilia drives multiscale mucus swirls in model human bronchial epithelium **8**(1), 2447 <https://doi.org/10.1038/s41598-018-20882-4>
- [19] Loiseau, E., Gsell, S., Nommick, A., Jomard, C., Gras, D., Chanez, P., D’Ortona, U., Kodjabachian, L., Favier, J., Viallat, A.: Active mucus–cilia hydrodynamic coupling drives self-organization of human bronchial epithelium **16**(11), 1158–1164 <https://doi.org/10.1038/s41567-020-0980-z>
- [20] Wan, K.Y., Goldstein, R.E.: Coordinated beating of algal flagella is mediated by basal coupling **113**(20) <https://doi.org/10.1073/pnas.1518527113>
- [21] Chakrabarti, B., Saintillan, D.: Hydrodynamic synchronization of spontaneously beating filaments **123**(20), 208101 <https://doi.org/10.1103/PhysRevLett.123.208101>
- [22] Grober, D., Dhar, T., Saintillan, D., Palacci, J.: The hydrodynamic torque dipole from rotary bacterial flagella powers symmetric discs <https://doi.org/10.1038/s41567-026-03189-4>
- [23] Vutukuri, H.R., Hoore, M., Abaurrea-Velasco, C., Van Buren, L., Dutto, A., Auth, T., Fedosov, D.A., Gompper, G., Vermant, J.: Active particles induce large shape deformations in giant lipid vesicles **586**(7827), 52–56 <https://doi.org/10.1038/s41586-020-2730-x> . Accessed 2026-05-18
- [24] Boudet, J.F., Lintuvuori, J., Lacouture, C., Barois, T., Deblais, A., Xie, K., Cassagnere, S., Tregon, B., Brückner, D.B., Baret, J.C., Kellay, H.: From collections of independent, mindless robots to flexible, mobile, and directional superstructures **6**(56), 0272 <https://doi.org/10.1126/scirobotics.abd0272> . Accessed 2026-05-18
- [25] Prasad, M., Obana, N., Lin, S.-Z., Zhao, S., Sakai, K., Blanch-Mercader, C., Prost, J., Nomura, N., Rupprecht, J.-F., Fattaccioli, J., Utada, A.S.: *Alcanivorax borkumensis* biofilms enhance oil degradation by interfacial tubulation **381**(6659), 748–753 <https://doi.org/10.1126/science.adf3345> . Accessed 2026-07-10
- [26] Liu, C., Cao, D., Liu, S., Wu, Y.: Nonequilibrium dynamics of membraneless active droplets. arXiv. <https://doi.org/10.48550/arXiv.2511.04181> . <http://arxiv.org/abs/2511.04181> Accessed 2026-07-10
- [27] Balboa Usabiaga, F., Kallemov, B., Delmotte, B., Bhalla, A., Griffith, B., Donev, A.: Hydrodynamics of suspensions of passive and active rigid particles: a rigid multiblob approach **11**(2), 217–296 <https://doi.org/10.2140/camcos.2016.11.217>
- [28] Balboa Usabiaga, F., Delmotte, B., Donev, A.: Brownian dynamics of confined suspensions of active microrollers **146**(13), 134104 <https://doi.org/10.1063/1.4979494>
- [29] Swan, J.W., Brady, J.F.: Simulation of hydrodynamically interacting particles near a no-slip boundary **19**(11), 113306 <https://doi.org/10.1063/1.2803837>
- [30] Dewangan, N.K., Conrad, J.C.: Rotating oil droplets driven by motile bacteria at interfaces. *Soft Matter* **15**(45), 9368–9375 (2019) <https://doi.org/10.1039/C9SM01570A>
- [31] Stone, H.A., Samuel, A.D.T.: Propulsion of microorganisms by surface distortions **77**(19), 4102–4104 <https://doi.org/10.1038/39131a>

- [32] Pak, O.S., Lauga, E.: Generalized squirming motion of a sphere **88**(1), 1–28 <https://doi.org/10.1007/s10665-014-9690-9> . Accessed 2023-10-27
- [33] Ramos, G., Cordero, M.L., Soto, R.: Bacteria driving droplets. *Soft Matter* **16**(5), 1359–1365 (2020) <https://doi.org/10.1039/C9SM01839E> . Publisher: The Royal Society of Chemistry
- [34] Altindal, T., Chattopadhyay, S., Wu, X.-L.: Bacterial chemotaxis in an optical trap **6**(4), 18231 <https://doi.org/10.1371/journal.pone.0018231> . Publisher: Public Library of Science
- [35] Brumley, D.R., Bruot, N., Kotar, J., Goldstein, R.E., Cicuta, P., Polin, M.: Long-range interactions, wobbles, and phase defects in chains of model cilia **1**(8), 081201 <https://doi.org/10.1103/PhysRevFluids.1.081201>
- [36] Lenz, P., Ryskin, A.: Collective effects in ciliar arrays **3**(4), 285–294 <https://doi.org/10.1088/1478-3975/3/4/006>
- [37] Niedermayer, T., Eckhardt, B., Lenz, P.: Synchronization, phase locking, and metachronal wave formation in ciliary chains **18**(3), 037128 <https://doi.org/10.1063/1.2956984>
- [38] Brumley, D.R., Polin, M., Pedley, T.J., Goldstein, R.E.: Metachronal waves in the flagellar beating of *Volvox* and their hydrodynamic origin **12**(108), 20141358 <https://doi.org/10.1098/rsif.2014.1358>
- [39] Sprinkle, B., Balboa Usabiaga, F., Patankar, N.A., Donev, A.: Large scale brownian dynamics of confined suspensions of rigid particles **147**(24), 244103 <https://doi.org/10.1063/1.5003833>
- [40] Squires, T.M., Mason, T.G.: Fluid mechanics of microrheology **42**(1), 413–438 <https://doi.org/10.1146/annurev-fluid-121108-145608>
- [41] Blake, J.R., Sleight, M.A.: Mechanics of ciliary locomotion. *Biological Reviews* **49**(1), 85–125 (1974)
- [42] Daddi-Moussa-Ider, A., Löwen, H., Liebchen, B.: Hydrodynamics can determine the optimal route for microswimmer navigation **4**(1), 1–11 <https://doi.org/10.1038/s42005-021-00522-6> . Publisher: Nature Publishing Group
- [43] Alas, C., Powers, T.R., Kuriabova, T.: Swimming of microorganisms in quasi-two-dimensional membranes **911**, 35 <https://doi.org/10.1017/jfm.2020.1043>
- [44] Qian, B., Jiang, H., Gagnon, D.A., Breuer, K.S., Powers, T.R.: Minimal model for synchronization induced by hydrodynamic interactions **80**(6), 061919 <https://doi.org/10.1103/PhysRevE.80.061919>
- [45] Blake, J.R.: A spherical envelope approach to ciliary propulsion **46**(1), 199–208 <https://doi.org/10.1017/S002211207100048X>
- [46] Taylor, M.E.: Partial Differential Equations. II, Qualitative Studies of Linear Equations, Third edition. edn. Applied Mathematical Sciences Series ; Volume 116. Springer. Publication Title: Partial Differential Equations. II, Qualitative Studies of Linear Equations
- [47] Pedley, T.J., Brumley, D.R., Goldstein, R.E.: Squirmers with swirl: a model for *Volvox* swimming **798**, 165–186 <https://doi.org/10.1017/jfm.2016.306>
- [48] Happel, J., Brenner, H.: Low Reynolds Number Hydrodynamics: with Special Applications to Particulate Media. Mechanics of fluids and transport processes, vol. 1. Springer. <https://doi.org/10.1007/978-94-009-8352-6> . <https://link.springer.com/10.1007/978-94-009-8352-6>
- [49] Green, C.C., Marshall, J.S.: Green’s function for the Laplace–Beltrami operator on a toroidal surface. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* **469**(2149), 20120479 (2013) <https://doi.org/10.1098/rspa.2012.0479>